

Multi-Area Asynchronous Grid Operation with Frequency Reserve Sharing

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Abstract—Frequency security of low-inertia power systems has become an increasing concern due to the high penetration of inverter-based resources. High-voltage direct current (HVDC) is a novel option for frequency regulation in multi-area asynchronous grids, and our focus is specifically on the primary frequency reserve sharing through HVDC. In this paper, we first develop a system frequency response model for multi-area asynchronous grids from which the frequency constraints can be extracted by an approximation algorithm. A frequency-constrained unit commitment model is then established, and an adaptive approach is introduced to modify the constraint bounds in order to fully ensure frequency security. Case study finally validates that the proposed method can maintain the frequency security of multi-area asynchronous grids as well as exploit the operational benefits of HVDC frequency regulation.

Index Terms—HVDC, frequency regulation, low inertia, unit commitment, inverter-based resource, renewable energy

I. INTRODUCTION

HIGH-voltage direct current (HVDC) technology is popular in transmission systems for long-distance transmission. Owing to its technical and economic benefits, HVDC is versatile to support asynchronous AC network interconnection [1], network flexibility [2], short-circuit current reduction [3], congestion management [4], and inter-regional coordinated dispatch [5]. In 2017, the global HVDC capacity reached 277 GW [6]. The scale of the global HVDC market is projected to have a compound average growth rate of 11% from 2020 to 2025 as well [7].

Using HVDC links to connect asynchronous AC grids is widespread around the world and multi-area asynchronous grids (MAG) have already existed in real-world implementation. In China, the Yunnan grid and the southern interconnection grid have been asynchronously connected to eliminate stability risks of the parallel AC/DC transmission system in case of DC line faults [8]. The flexible HVDC links are able to address the angle instability resulting from tie-line

overloads [9]. In the U.S., the Eastern Interconnection is tied to the Western Interconnection by six DC links and to the Texas Interconnection by two DC links. Reference [10] showed that the benefit-to-cost ratio of increasing HVDC capacity between the Eastern and the Western Interconnection reached 2.5.

These HVDC systems are critical resources to support the frequency security of MAG which is expected to integrate a large number of power electronic inverters in the future [11]. Maintaining adequate primary frequency reserves is important to keep the frequency nadir within a secure range. In this context, various HVDC controllers have been developed for frequency support, and the general idea is to add a frequency feedback loop in calculating the active power reference points. In [12], a frequency limit control strategy was proposed for the sending-end grid, and the performance was validated through practical system testing. An adaptive droop control strategy for multi-terminal DC (MTDC) was designed in [13] to facilitate mutual frequency support among asynchronous AC grids. In [14], a frequency control structure for voltage source converter based MTDC (VSC-MTDC) was proposed, and an overall frequency reference signal was adopted to smooth the DC voltage profile. In [15], a frequency control strategy for VSC-MTDC systems to enable the sharing of both primary and secondary frequency reserves was proposed. A distributed frequency controller for MTDC that only needs local and neighboring information was designed in [16].

Frequency security can be evaluated in the operation stage by developing precise frequency security constraints. To do this, some efforts were focused on extracting explicit constraint expressions from nonlinear system frequency response (SFR). Note that the system frequency response characteristics are mainly determined by the on/off statuses of the generators, so the frequency security constraints are commonly enforced in the unit commitment problem, namely frequency-constrained unit commitment (FCUC). In [17], the minimum frequency was calculated by the SFR model, represented by a piecewise linearization technique, and integrated into the FCUC problem. In [18], the expression of frequency security constraints was reformulated as frequency security margin, and the frequency regulation from variable renewable energy was considered. In [19], a stochastic FCUC model was developed under a simplified increasing-ramp assumption of the governor droop control. A stochastic unit commitment model with more precise frequency security constraints extracted from the SFR model was further proposed in [20]. In [21], a frequency-constrained scheduling approach was proposed, with accurate modeling of wind turbine capacity and wake effect. In [22], the fast

This work was supported by the National Natural Science Foundation of China under Grant U22B6007 and 52122706. (Corresponding author: Haiwang Zhong)

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frequency response of battery energy storage was considered in the FCUC model. In addition, data-driven methods were also applied by some papers to extract frequency constraints. In [23], deep neural networks were trained to represent the frequency constraints and converted to a mixed-integer linear formulation. In [24], optimal classification trees were proposed to extract the frequency constraints from a detailed SFR model considering control dead band and saturation. A comprehensive review of the frequency constraint linearization methods was provided in [25].

Several papers have studied the frequency-constrained operation for asynchronous grids with HVDC frequency support. In [9], a frequency-constrained optimal power flow model with HVDC emergency power support was proposed. In [26], an FCUC model for asynchronous AC systems was proposed, where HVDC supplementary power control for primary frequency exchange was considered. In [27]–[29], the FCUC model was developed under a simplified assumption that the HVDC power flow has a step change in the moment of contingency and then recovers to a post-contingency value with a fixed rate. In the aforementioned papers concerning the frequency-constrained operation for asynchronous grids [9], [26]–[29], the studied system consists of two AC grids connected with each other by HVDC links, and how to extend their methods to a more generalized scenario with more interconnected AC grids was not explained. When multiple AC grids are interconnected and provide frequency support for each other, the system frequency dynamics become much more complex. The frequencies of these AC grids are coupled by the frequency regulation characteristics of the HVDC links. When power imbalance occurs in one AC grid, all the interconnected AC grids will provide a certain level of frequency support to the system and their frequencies will be influenced. With such complex mutual influence, the frequency dynamics of multi-area asynchronous grids has not been quantitatively modeled and analyzed in the existing work.

Therefore, it is necessary to study the high-precision frequency response model of asynchronous AC grids under complex interactions caused by HVDC frequency support. This paper develops a system frequency response model for MAG by analytical deduction and proposes an approximation algorithm to extract frequency constraints. An FCUC model for multi-area asynchronous grids (FCUC-MAG) is proposed. The main contributions are summarized below:

- 1) A system frequency response model for multi-area asynchronous grids (SFR-MAG) considering the frequency regulation of HVDC is developed. Different from the existing frequency response model for one AC grid, the SFR-MAG model can appropriately consider the mutual influences on frequency response characteristics caused by HVDC frequency regulation and describe the relationship between the power disturbances and the resulting frequency deviations of all the asynchronously interconnected AC grids.
- 2) A frequency constraint extraction algorithm is specially designed based on the structure and the dominant terms of the SFR-MAG model. An approximate solution of the SFR-MAG model is first obtained based on a parameter

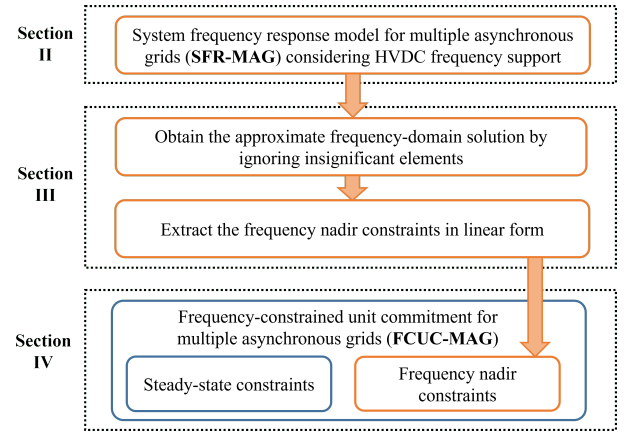


Fig. 1. The framework of this work.

aggregation method, which can effectively decrease the dimension of the subsequent linearization process. The frequency nadir constraints are then extracted using convex piecewise-linear regression.

- 3) A frequency-constrained unit commitment model for multi-area asynchronous grids (FCUC-MAG) is proposed. A frequency constraint modification approach is designed to guarantee the frequency security limit can be strictly satisfied even if there are errors in the fitted expression of the frequency constraint.

The framework of this paper is shown in Fig. 1. Section II will establish a frequency response model considering the frequency response characteristics of AC grids and interconnecting DC lines. In Section III, this model will be further piecewise-linear approximated the frequency nadir expressions. The FCUC-MAG model will be described later in Section IV.

II. SYSTEM FREQUENCY RESPONSE MODEL FOR MULTI-AREA ASYNCHRONOUS GRIDS

In this section, the system frequency response model for multi-area asynchronous grids is developed. The system frequency response of a single AC grid and the frequency regulation characteristics of HVDC are introduced. Considering the mutual influences on frequency response characteristics among neighboring AC grids, the system frequency response model is derived in a matrix form to describe the relationship between the power disturbances and the resulting frequency deviations of all the asynchronously interconnected AC grids.

A. An illustration of MAG

Multi-area asynchronous grids (MAG) refer to a group of AC grids interconnected by HVDC and MTDC. Fig. 2 gives an example of multi-area asynchronous grids, where AC Grids 1,2,3 are interconnected by an MTDC system and AC Grid 4 is connected to AC Grid 3 by a point-to-point HVDC. This structure is generally designed to take advantage of the high operational controllability of HVDC, as well as decrease long-distance transmission loss. The main applications include interconnecting major AC grids and transmitting

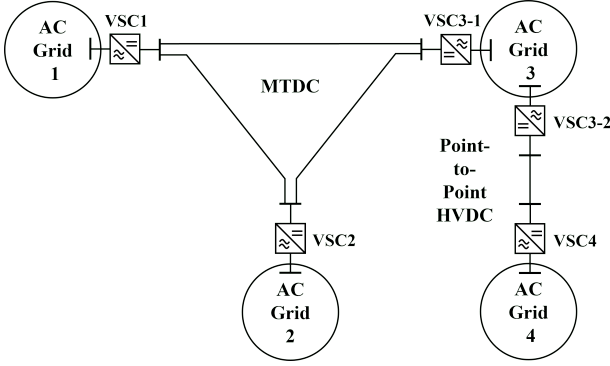


Fig. 2. An example of multi-area asynchronous grids.

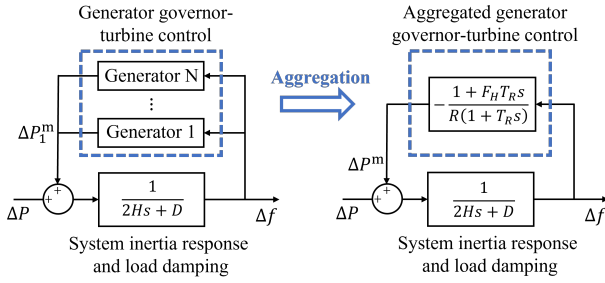


Fig. 3. The block diagrams of the system swing equation and the ASFR model.

distant renewable energy resources. In traditional scheduling schemes, the active power set points of DC systems are fixed. Each AC grid operates independently, taking the power flows of DC systems as boundary conditions. By exploiting the frequency support capability of HVDC, MAG enjoy the benefits of sharing primary frequency reserves, improving frequency security margin, and reducing operational costs. After introducing HVDC frequency control, the frequency deviation of one AC grid influences the power flows of DC lines that are directly connected to it, hence impacting the neighboring AC grids. Under such mutual influences, the frequency response characteristics of each AC grid are complex.

B. Frequency Response of a Single Grid

The frequency response of a single AC grid can be described by the system swing equation in (1), where H is the equivalent system inertia, D is the system load damping ratio, $\Delta f(s)$ is the frequency deviation, $\Delta P(s)$ is the imbalanced electrical power, and $\Delta P_g^m(s)$ is the mechanical power adjustment of generator g . In [30], an aggregation method was proposed to aggregate the multi-machine model into a single-machine model, i.e. the aggregated system frequency response (ASFR) model, as shown in 3. The upper block reflects the aggregated generator governor control and the lower block reflects the system inertia response and load damping. In this work, the ASFR model is developed based on a rudimentary governor-turbine model, as given in (2), where the transfer function $G(s)$ is defined as the ratio of $\Delta f(s)$ to $\Delta P(s)$, R is the equivalent governor droop value, F_H is the high-pressure turbine fraction, and T_R is the reheat time constant. The equivalent parameters

of the ASFR model can be calculated by (3)-(6), where K_g , R_g , F_g , H_g , T_g , $P_g^{\max}$, and u_g^t are the mechanical power gain, the governor droop, the high-pressure turbine fraction, the inertia constant, the reheat time, the maximum active power capacity, and the on/off status of generator g . $\mathcal{G}$ denotes the set of all generators. d_{sys}^t is the total demand of the AC system at time t . More sophisticated governor-turbine models, e.g. IEEE G1 Steam Turbine model, involve other control elements, such as lead-lag compensator and steam chest time constant. However, the reheat time constant is typically significantly greater than the time constants in the other elements [31]. Based on the dominant pole approximation, we assume that the system frequency response is dominated by the slowest time constant, and those elements with much smaller time constant are ignored. Moreover, the nonlinear elements such as the frequency deadband and saturation are also ignored to derive an analytical expression of the ASFR model.

$$(2Hs + D) \Delta f(s) = \Delta P(s) - \sum \Delta P_g^m(s) \quad (1)$$

$$G(s) = \frac{\Delta f(s)}{\Delta P(s)} = \frac{1}{2Hs + D + \frac{1 + F_H T_R s}{R(1 + T_R s)}} \quad (2)$$

$$\frac{1}{R} = \sum_{g \in \mathcal{G}} \frac{K_g}{R_g} \frac{P_g^{\max} u_g^t}{d_{\text{sys}}^t} \quad (3)$$

$$\frac{F_H}{R} = \sum_{g \in \mathcal{G}} \frac{K_g F_g}{R_g} \frac{P_g^{\max} u_g^t}{d_{\text{sys}}^t} \quad (4)$$

$$H = \frac{\sum_{g \in \mathcal{G}} H_g P_g^{\max} u_g^t}{d_{\text{sys}}^t} \quad (5)$$

$$T_R = \sum_{g \in \mathcal{G}} \frac{K_g}{R_g} \frac{P_g^{\max} u_g^t}{d_{\text{sys}}^t} T_g \quad (6)$$

In most existing literature, the reheat time T_R is assumed an identical value due to its insignificant impact. The load damping ratio D is proportional to the total load. The parameters in the swing equation (1) are usually normalized by the total load, so that D can be treated as a constant. Therefore, $\frac{1}{R}$, $\frac{F_H}{R}$, and H are usually treated as the inputs of the ASFR model, which can be expressed as the weighted sums of u_g^t .

C. Frequency Regulation of HVDC

In this work, we assume applying the constant DC voltage control as the basic control mode of the HVDC converters. For a point-to-point HVDC system, one converter is configured with a DC voltage controller and the other is configured with a DC power controller, as shown in Fig. 4. In order to enable HVDC frequency regulation, a supplementary frequency controller is added to modify the DC power reference according to the frequency deviation. We assume the HVDC system provides frequency support for both asynchronous grids that it connects, and a frequency droop control strategy is adopted, as shown in (7), where $\Delta P_{DC}(s)$ is the active power adjustment of the HVDC line. $\Delta f^i(s)$ and $\Delta f^j(s)$ are the frequency deviations of the sending-end and receiving-end

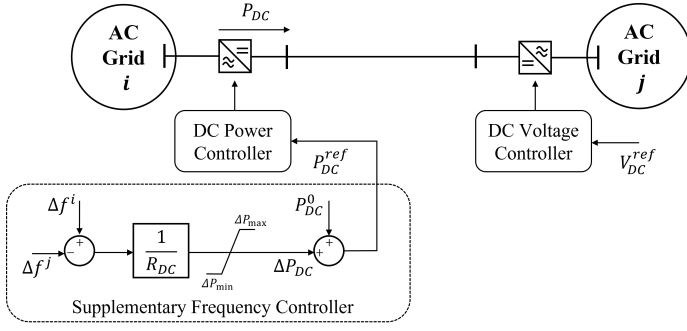


Fig. 4. HVDC controller configuration.

grids respectively, and R_{DC} is the droop coefficient. When the sending-end frequency drops, the HVDC line decreases the active power drawn from the sending-end grid, and vice versa. By this means, the two AC grids connected by HVDC can provide frequency support for each other, so that the frequency deviation of each grid is maintained within a smaller range. The DC power reference modification ΔP_{DC} is then actuated by an inner current controller of the DC power controller. The time constant of the inner current controller is relatively small (typically several milliseconds [14]), and thus is ignored.

$$\Delta P_{DC}(s) = \frac{1}{R_{DC}} (\Delta f^i(s) - \Delta f^j(s)) \quad (7)$$

Assume Grid i and j are connected by several HVDC lines, then the total adjustment of power interchange between these two grids $\Delta P^{i,j}(s)$ is expressed in (8). We refer to $G^{i,j}(s)$ as the mutual transfer function between AC Grid i and j , which indicates the strength of the mutual frequency support through HVDC. It can be calculated by (9), where $\mathcal{L}_{DC}$ denotes the set of HVDC lines between Grid i and j .

$$\Delta P^{i,j}(s) = \frac{1}{G^{i,j}(s)} (\Delta f^i(s) - \Delta f^j(s)) \quad (8)$$

$$\frac{1}{G^{i,j}(s)} = \sum_{l \in \mathcal{L}_{DC}(i,j)} \frac{1}{R_{DC}^l} \quad (9)$$

For an MTDC system, we assume the frequency regulation strategy in [14] is applied. The active power adjustment of the converter connected to AC Grid i $\Delta P_{DC}^i(s)$ is given by (10), where R_{DC}^i is the droop coefficient of the converter. The global frequency reference signal of the MTDC system $\Delta f^{ref}(s)$ is calculated by the weighted average of frequencies at all converter AC terminals, which is given by (11). The adjustment of power interchange between AC Grid i and j can be expressed in the same way as (8), with $\frac{1}{G^{i,j}(s)}$ given by (12). When an AC grid is connected with both MTDC and point-to-point HVDC, we need to add (9) and (12) to obtain the expression of $\frac{1}{G^{i,j}(s)}$.

$$\Delta P_{DC}^i(s) = \frac{1}{R_{DC}^i} (\Delta f^i(s) - \Delta f^{ref}(s)) \quad (10)$$

$$\Delta f^{ref}(s) = \frac{\sum \frac{\Delta f^k(s)}{R_{DC}^k}}{\sum \frac{1}{R_{DC}^k}} \quad (11)$$

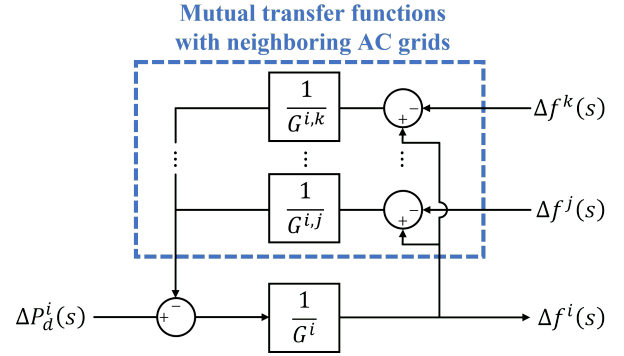


Fig. 5. The block diagram of the SFR-MAG model.

$$\frac{1}{G^{i,j}(s)} = \frac{1}{R_{DC}^i R_{DC}^j \sum \frac{1}{R_{DC}^k}} \quad (12)$$

D. Frequency Response of Multi-area Asynchronous Grids

Fig. 5 is the block diagram of the SFR-MAG model. Assume a power disturbance $\Delta P_d^i(s)$ occurs in AC Grid i . The total imbalanced power of AC Grid i is the sum of the initial power disturbance within the grid $\Delta P_d^i(s)$ and the changes of the active power of HVDC lines, as shown in (13), where $\mathcal{N}(i)$ denotes the set of neighboring grids that are asynchronously connected to Grid i by HVDC.

$$\Delta P^i(s) = \Delta P_d^i(s) - \sum_{j \in \mathcal{N}(i)} \Delta P^{i,j}(s) \quad (13)$$

By substituting $\Delta P^{i,j}(s)$ with the detailed expression in (8), the total imbalanced power $\Delta P^i(s)$ is obtained as shown in (14).

$$\Delta P^i(s) = \Delta P_d^i(s) - \sum_{j \in \mathcal{N}(i)} \frac{\Delta f^i(s)}{G^{i,j}(s)} + \sum_{j \in \mathcal{N}(i)} \frac{\Delta f^j(s)}{G^{i,j}(s)} \quad (14)$$

The frequency response of AC Grid i is described by (15), where $G^i(s)$ is its original transfer function when HVDC frequency regulation is not considered. The expression of $G^i(s)$ is given in (16), where the superscript i of each parameter denotes AC Grid i .

$$\Delta P^i(s) = \frac{\Delta f^i(s)}{G^i(s)} \quad (15)$$

$$G^i(s) = \frac{1}{2H^i s + D^i + \frac{1 + F_H^i T_R^i s}{R^i (1 + T_R^i s)}} \quad (16)$$

Combining (14) and (15), we can obtain (17), which describes how the frequency response of Grid i is impacted by its initial power disturbance and the frequency deviations of neighboring grids. Note that (s) is omitted for simplicity.

$$\left(\frac{1}{G^i} + \sum_{j \in \mathcal{N}(i)} \frac{1}{G^{i,j}} \right) \Delta f^i - \sum_{j \in \mathcal{N}(i)} \frac{\Delta f^j}{G^{i,j}} = \Delta P_d^i \quad (17)$$

By applying (17) to each AC grid and joining these equations together, we can obtain the matrix form of the SFR-MAG

model, as shown in (18). The vector $\Delta \mathbf{f}$ and $\Delta \mathbf{P}_d$ denote the frequency deviations and the local power disturbances of all the interconnected AC grids. It is a multiple-input multiple-output system model in the frequency domain. According to (17), each element of $\mathbf{W}$ is given in (19)-(21).

$$\mathbf{W} \Delta \mathbf{f} = \Delta \mathbf{P}_d \quad (18)$$

$$W_{ii} = \frac{1}{G^i} + \sum_{j \in \mathcal{N}(i)} \frac{1}{G^{i,j}} \quad (19)$$

$$W_{ij} = -\frac{1}{G^{i,j}}, j \in \mathcal{N}(i) \quad (20)$$

$$W_{ij} = 0, j \notin \mathcal{N}(i) \quad (21)$$

III. EXTRACT FREQUENCY NADIR CONSTRAINTS

In this section, a frequency constraint extraction algorithm is developed. An approximate solution of the SFR-MAG model in (18) is obtained based on a parameter aggregation method, which can effectively decrease the dimension of the subsequent linearization process. Using the frequency-domain solution obtained above, data are generated by calculating system step response. A convex piecewise-linear regression algorithm is applied to obtain a linear expression of the frequency nadir.

A. Approximate Solution by Parameter Aggregation

Consider a power disturbance ΔP_d^1 happens in AC Grid 1, then the power disturbance vector is shown in (22), where $\mathcal{A}$ denotes the set of all AC grids and $|\mathcal{A}|$ denotes the number of AC grids. In most cases, the power disturbance is a step function, i.e. $\Delta P_d^1(s) = \frac{\Delta P_d^1}{s}$, where ΔP_d^1 is the magnitude of the power disturbance. If we want to know the exact frequency deviation of each grid, we need to solve (18) by calculating the inverse of $\mathbf{W}$, which is computationally expensive. Moreover, the exact solution is related to the parameters of all the asynchronous grids, which will cause intractability in the subsequent step of extracting explicit frequency nadir constraints. Therefore, we try to obtain an approximate solution with an explicit expression by ignoring some insignificant elements.

$$\Delta \mathbf{P}_d = \begin{bmatrix} \Delta P_d^1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{|\mathcal{A}| \times 1} \quad (22)$$

It is assumed that the frequency regulation capability of HVDC is significantly smaller than that of each regional AC grid, i.e., the magnitude of $\frac{1}{G^{i,j}}$ is significantly smaller than the magnitude of $\frac{1}{G^i}$. This can be explained by the fact that the frequency regulation capability is a certain proportion (usually around 10%) of the rated capacity for both HVDC and generators. Therefore, the total frequency regulation capability is mainly determined by the total installed capacity. In practice, the total installed capacity of the HVDC interconnections is usually much smaller than the total installed generation capacity of a regional AC grid. In the case of the Nordic synchronous area (Norway, Sweden, Finland and Eastern

Denmark) [32], the total installed generation capacity is 97.2 GW and the total demand ranges from 26.4 to 60.1 GW, but the HVDC interconnections to neighborhood areas are 8.3 GW altogether. Therefore, we regard the magnitude of $\frac{1}{G^{i,j}}$ as the first-order infinitesimal. In the rest of this section, we only concern the first-order infinitesimal and the second-order infinitesimal, and the terms with less magnitude are ignored. In order to derive the frequency response of Grid 1 (where the initial power disturbance happens), (18) is split into (23). Since only the first row is of interest, we obtain (24) by eliminating the other rows.

$$\begin{bmatrix} W_{11} & \mathbf{W}_{1k} \\ \mathbf{W}_{1k}^T & \mathbf{W}_{kk} \end{bmatrix} \Delta \mathbf{f} = \Delta \mathbf{P}_d \quad (23)$$

$$(W_{11} - \mathbf{W}_{1k} \mathbf{W}_{kk}^{-1} \mathbf{W}_{1k}^T) \Delta f^1 = \Delta P_d^1 \quad (24)$$

To derive Δf^1 , the most simplified way is to ignore $\mathbf{W}_{1k} \mathbf{W}_{kk}^{-1} \mathbf{W}_{1k}^T$.

$$\Delta f^1 \approx \frac{\Delta P_d^1}{W_{11}} \quad (25)$$

This approximation method is referred to as the simplified method. However, the simplified method can cause a relatively large error, which will be further presented in Section V.

Note that each element of $\mathbf{W}_{1k}$ is of the same magnitude as $\frac{1}{G^{i,j}}$. While calculating $\mathbf{W}_{1k} \mathbf{W}_{kk}^{-1} \mathbf{W}_{1k}^T$, we only concern the elements of the same magnitude as $\frac{1}{G^i}$ in $\mathbf{W}_{kk}$, i.e., the diagonal elements of $\mathbf{W}_{kk}$. The elements with smaller magnitude in $\mathbf{W}_{kk}^{-1}$ is corresponding to the third-order infinitesimals in $\mathbf{W}_{1k} \mathbf{W}_{kk}^{-1} \mathbf{W}_{1k}^T$. This leads to the approximation in (26) and (27). Equation (24) is simplified into (28).

$$\mathbf{W}_{kk}^{-1} \approx \text{diag} \left(\frac{1}{W_{22}}, \dots, \frac{1}{W_{NN}} \right) \quad (26)$$

$$\mathbf{W}_{1k} \mathbf{W}_{kk}^{-1} \mathbf{W}_{1k}^T \approx \sum_{i=2}^N \frac{W_{1i} W_{i1}}{W_{ii}} \quad (27)$$

$$\left(W_{11} - \sum_{i=2}^N \frac{W_{1i} W_{i1}}{W_{ii}} \right) \Delta f^1 = \Delta P_d^1 \quad (28)$$

Note that $W_{1i}, W_{i1} \neq 0$ only for $i \in \mathcal{N}(1)$, we can obtain the expression of Δf^1 as in (29). The physical explanation of this step of approximation is that only the influence of the directly adjacent AC grids is considered.

$$\Delta f^1 = \frac{\Delta P_d^1}{\left(W_{11} - \sum_{i \in \mathcal{N}(1)} \frac{W_{1i} W_{i1}}{W_{ii}} \right)} \quad (29)$$

The approximate frequency-domain solution in (29) includes $\sum_{i \in \mathcal{N}(1)} \frac{W_{1i} W_{i1}}{W_{ii}}$, which is a summation term over all the neighboring grids. It should be noted that in the case of multi-area asynchronous grids, the parameters of each grid are normalized by the summation of loads over all the grids, and D^i becomes a variable. Therefore, for each neighboring grid, there are four variables we need to consider: H^i , $\frac{1}{R^i}$, $\frac{F_H^i}{R^i}$, and D^i . The total number of variables is $4|\mathcal{N}(1)|$, where $|\mathcal{N}(1)|$ is the number of grids directly adjacent to Grid 1. The number of variables is not fixed, and the subsequent step

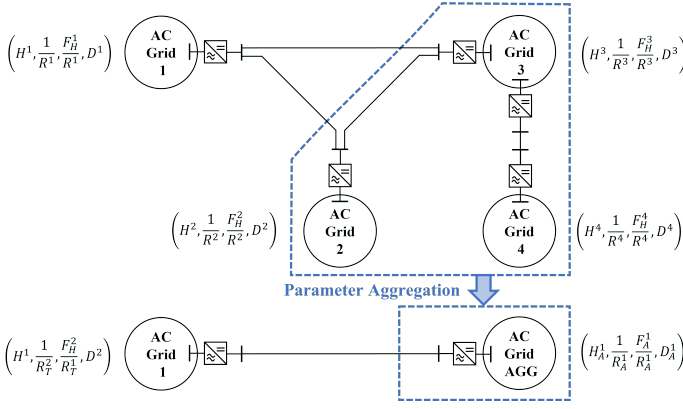


Fig. 6. Aggregation of neighboring AC grids.

of convex piecewise-linear regression may become intractable as the number of adjacent grids increases.

To handle this problem, a parameter aggregation technique is designed to further simplify the expression and reduce the dimension of the variables. Similar to [30], the main idea is to aggregate the parameters of the rest AC grids into one set of parameters, as shown in Fig. 6. The aggregated parameters are calculated in a reciprocal manner, as shown in (30)-(33):

$$\frac{1}{H_A^1} = \sum_{i \in \mathcal{N}(1)} \frac{1}{G^{1,i} G^{i,1} H^i} \quad (30)$$

$$R_A^1 = \sum_{i \in \mathcal{N}(1)} \frac{R^i}{G^{1,i} G^{i,1}} \quad (31)$$

$$\frac{R_A^1}{F_A^1} = \sum_{i \in \mathcal{N}(1)} \frac{R^i}{G^{1,i} G^{i,1} F_H^i} \quad (32)$$

$$\frac{1}{D_A^1} = \sum_{i \in \mathcal{N}(1)} \frac{1}{G^{1,i} G^{i,1} \left(D^i + \sum_{j \in \mathcal{N}(i)} \frac{1}{G^{i,j}} \right)} \quad (33)$$

The summation term $\frac{W_{1i}W_{i1}}{W_{ii}}$ is approximately substituted using the aggregated parameters:

$$\sum_{i \in \mathcal{N}(1)} \frac{W_{1i}W_{i1}}{W_{ii}} \approx \frac{1}{2H_A^1 s + D_A^1 + \frac{1 + F_A^1 T_A^1 s}{R_A^1 (1 + T_A^1 s)}} \quad (34)$$

B. Convex Piecewise-Linear Regression of Frequency Nadir Expression

Let $\mathbf{X}^1 = [H^1; \frac{1}{R^1}; \frac{F_H^1}{R^1}; D^1; \frac{1}{H^1}; R_A^1; \frac{R_A^1}{F_A^1}; \frac{1}{D_A^1}]$ refer to the parameter vector of Grid 1. After the above-mentioned parameter aggregation method, the frequency response of Grid 1 can be expressed as a function of $\mathbf{X}^1$:

$$\frac{\Delta f^1(s)}{\Delta P_d^1(s)} = g(\mathbf{X}^1) \quad (35)$$

Assuming a step power disturbance $\Delta P_d^1(s) = \frac{\Delta P^1}{s}$, we can obtain the time-domain solution of Δf^1 by applying inverse Laplace transform:

$$\Delta f^1(t) = \mathcal{L}^{-1} \left[\frac{g(\mathbf{X}^1)}{s} \right] \Delta P^1 \quad (36)$$

Let $h(\mathbf{X}^1) = \max \left(\mathcal{L}^{-1} \left[\frac{g(\mathbf{X}^1)}{s} \right] \right)$ denote the maximum value of the time-domain solution and $f_{\max}^1$ denotes the predefined frequency security limit of Grid 1, then the frequency nadir constraint can be expressed as (37).

$$\frac{1}{h(\mathbf{X}^1)} \geq \frac{\Delta P^1}{f_{\max}^1} \quad (37)$$

Convex piecewise-linear regression is applied to transform the frequency nadir constraint (37) into a linear form. Assume a dataset $(\mathbf{X}_n^1, y_n^1)$, $n = 1, \dots, N_{\text{data}}$ is generated by the function $\frac{1}{h(\mathbf{X}^1)}$. The detailed process of dataset generation is given in Appendix A. By solving (38)-(39), we can obtain K_{pwl} sets of linear coefficients and intercepts $(\alpha_{(1)}, \beta_{(1)}), \dots, (\alpha_{(K_{\text{pwl}})}, \beta_{(K_{\text{pwl}})})$. The linearized frequency nadir constraint of Grid 1 can be expressed as (40). For other grids, the procedure to generate the frequency nadir constraints is the same.

$$\min \sum_{n=1}^{N_{\text{data}}} (\hat{y}_n - y_n)^2 \quad (38)$$

$$\hat{y}_n = \min \left\{ \alpha_{(1)}^T \mathbf{X}_n^1 + \beta_{(1)}, \dots, \alpha_{(K_{\text{pwl}})}^T \mathbf{X}_n^1 + \beta_{(K_{\text{pwl}})} \right\} \quad (39)$$

$$\alpha_{(k)}^T \mathbf{X}^1 + \beta_{(k)} \geq \frac{\Delta P^1}{\Delta f_{\max}^1}, k = 1, \dots, K_{\text{pwl}} \quad (40)$$

IV. FREQUENCY-CONSTRAINED UNIT COMMITMENT FOR MULTI-AREA ASYNCHRONOUS GRIDS

A. Optimization Model Formulation

The objective function of the FCUC-MAG is to minimize the operational costs within the time horizon Ω_t , including the generation cost, the start-up cost, and the shutdown cost, where P_g^t is the output of generator g at time t and $C_g(P_g^t)$ is the piecewise-linear cost of generator g :

$$\min \sum_{t \in \Omega_t} \sum_{g \in \mathcal{G}} C_g(P_g^t) + SU_g^t + SD_g^t \quad (41)$$

The constraints of the FCUC-MAG model include:

1) *Start-up Cost and Shutdown Cost*: By satisfying (42) and (43), the auxiliary variable SU_g^t is greater than or equal to the actual start-up cost of generator g in time period t , where C_g^{SU} is the cost of one start-up. The equality holds at the optimal solution. Similar constraints on SD_g^t is shown in (44) and (45).

$$SU_g^t \geq 0 \quad (42)$$

$$SU_g^t \geq C_g^{SU} (u_g^t - u_g^{t-1}) \quad (43)$$

$$SD_g^t \geq 0 \quad (44)$$

$$SD_g^t \geq C_g^{SD} (u_g^{t-1} - u_g^t) \quad (45)$$

2) *Power Balance Constraints*: The power balance constraints with DC power flow model are shown in (46), where $P_p^t, P_{D,p}^t, \theta_p, \theta_q, X_{pq}$, and $P_{DC,p}^t$ denote the total power generation and demand on bus p , the voltage angle on bus p and q , the branch reactance between bus p and q , and the power injection of HVDC lines on bus p .

$$P_p^t - P_{D,p}^t = \sum_{q \in \mathcal{N}_{AC}(p)} \frac{\theta_p - \theta_q}{X_{pq}} + P_{DC,p}^t \quad (46)$$

The power balance constraints of HVDC lines are shown in (47), where the transmission losses are ignored. $P_{DC,p}$ and $P_{DC,q}$ denote the power injections at the sending end and receiving end of HVDC. $\mathcal{L}_{DC}(p, q)$ denotes the set of HVDC lines between bus p and q .

$$P_{DC,p}^t + P_{DC,q}^t = 0, \forall \mathcal{L}_{DC}(p, q) \quad (47)$$

3) *Generator Constraints*: Equation (48) limits the active power output of each generator.

$$P_g^{\min} u_g^t \leq P_g^t \leq P_g^{\max} u_g^t \quad (48)$$

Equation (49) and (50) are the upward and downward ramping constraints of each generator, which are expressed with big-M relaxation.

$$P_g^t - P_g^{t-1} \leq P_g^{\text{up}} + (2 - u_g^{t-1} - u_g^t) M \quad (49)$$

$$P_g^{t-1} - P_g^t \leq P_g^{\text{down}} + (2 - u_g^{t-1} - u_g^t) M \quad (50)$$

The minimum online time constraints are shown in (51) and (52), where T_g^{on} denotes the minimum online time of generator g . Equation (51) guarantees that once the generator starts up, it remains online for at least T_g^{on} . Equation (52) guarantees if the generator starts up in the final T_g^{on} time periods, it remains online until the end of the optimization horizon T .

$$\sum_{k=t}^{t+T_g^{\text{on}}-1} u_g^k \geq T_g^{\text{on}} (u_g^t - u_g^{t-1}), t = 1, \dots, T - T_g^{\text{on}} + 1 \quad (51)$$

$$\sum_{k=t}^T [u_g^k - (u_g^t - u_g^{t-1})] \geq 0, t = T - T_g^{\text{on}} + 2, \dots, T \quad (52)$$

Similarly, the minimum offline time constraints are given in (53) and (54).

$$\sum_{k=t}^{t+T_g^{\text{off}}-1} (1 - u_g^k) \geq T_g^{\text{off}} (u_g^{t-1} - u_g^t), t = 1, \dots, T - T_g^{\text{off}} + 1 \quad (53)$$

$$\sum_{k=t}^T [1 - u_g^k - (u_g^{t-1} - u_g^t)] \geq 0, t = T - T_g^{\text{off}} + 2, \dots, T \quad (54)$$

4) *Transmission Line Constraints*: The bidirectional power flow over each transmission line should not exceed its capacity. Equation (55) and (56) are the power flow constraints for AC lines and DC lines respectively.

$$-P_{AC,l}^{\max} \leq \frac{\theta_p - \theta_q}{X_l} \leq P_{AC,l}^{\max}, \forall l \in \mathcal{L}_{AC}(p, q) \quad (55)$$

$$-P_{DC,l}^{\max} \leq P_{DC,p}^t, P_{DC,q}^t \leq P_{DC,l}^{\max}, \forall l \in \mathcal{L}_{DC}(p, q) \quad (56)$$

5) *Reserve Capacity Constraints*: The generators need to provide enough reserve capacities for primary frequency regulation. Only upward reserve is considered for simplicity, as shown in (57), where $\mathcal{G}(i)$ denotes the set of generators in Grid i .

$$P_g^{\max} u_g^t - P_g^t \geq u_g^t \frac{K_g}{R_g} P_g^{\max} \Delta f_{\max}^i, \forall g \in \mathcal{G}(i) \quad (57)$$

To provide frequency support, HVDC lines also need to maintain enough reserve capacities:

$$P_{DC,l}^{\max} - \frac{K_{DC}^l P_{DC,l}^{\max} \Delta f_{\max}^i}{R_{DC}^l} \geq P_{DC,p}^t, P_{DC,q}^t, \forall l \in \mathcal{L}_{DC}(p, q) \quad (58)$$

6) *Frequency Nadir Constraints*: Each AC grid should satisfy the frequency nadir constraints expressed in the convex piecewise-linear form according to (40), where $\mathcal{A}$ denotes the set of all AC grids.

$$\alpha_{(k)}^T \mathbf{X}^{i,t} + \beta_{(k)} \geq \frac{\Delta P^{i,t}}{\Delta f_{\max}^i}, \forall k, \forall i \in \mathcal{A}, \forall t \in \Omega_t \quad (59)$$

Recall that $\mathbf{X}^{i,t}$ includes eight parameters. Four parameters of Grid i itself are calculated by:

$$H^{i,t} = \frac{\sum_{g \in \mathcal{G}(i)} H_g P_g^{\max} u_g^t}{d_{\text{sys}}^t} \quad (60)$$

$$\frac{1}{R^{i,t}} = \sum_{g \in \mathcal{G}(i)} \frac{K_g}{R_g} \frac{P_g^{\max} u_g^t}{d_{\text{sys}}^t} \quad (61)$$

$$\frac{F_H^{i,t}}{R^{i,t}} = \sum_{g \in \mathcal{G}(i)} \frac{K_g F_g}{R_g} \frac{P_g^{\max} u_g^t}{d_{\text{sys}}^t} \quad (62)$$

$$D^{i,t} = D \frac{d_{\text{sys}}^t}{d_{\text{sys}}^t} \quad (63)$$

Four aggregated parameters of neighboring grids are calculated by:

$$\frac{1}{H_A^{i,t}} = \sum_{j \in \mathcal{N}(i)} \frac{d_{\text{sys}}^t}{G^{i,j} G^{j,i} \sum_{g \in \mathcal{G}(j)} H_g P_g^{\max} u_g^t} \quad (64)$$

$$R_A^{i,t} = \sum_{j \in \mathcal{N}(i)} \frac{d_{\text{sys}}^t}{G^{i,j} G^{j,i} \sum_{g \in \mathcal{G}(j)} \frac{K_g}{R_g} P_g^{\max} u_g^t} \quad (65)$$

$$\frac{R_A^{i,t}}{F_A^{i,t}} = \sum_{j \in \mathcal{N}(i)} \frac{d_{\text{sys}}^t}{G^{i,j} G^{j,i} \sum_{g \in \mathcal{G}(j)} \frac{K_g F_g}{R_g} P_g^{\max} u_g^t} \quad (66)$$

$$\frac{1}{D_A^{i,t}} = \sum_{j \in \mathcal{N}(i)} \frac{1}{G^{i,j} G^{j,i} \left(D^{j,t} + \sum_{k \in \mathcal{N}(j)} \frac{1}{G^{j,k}} \right)} \quad (67)$$

It should be noted that (64)-(66) include fractional terms of binary variables u_g^t , making them non-linear constraints. A technique to transform such fractional terms of binary variables into linear constraints is applied, which is presented in the Appendix.

B. Frequency Constraint Modification

The frequency nadir constraints (59) are the results of approximation and regression, so errors are inevitable. Moreover, the calculation results of the ASFR model can also have errors compared with more detailed model. To guarantee the frequency security limit can be strictly satisfied in the presence of errors, a frequency constraint modification approach is designed. The main idea is to modify the frequency security limit based on the actual frequency violations and to solve the optimization model again with stricter frequency constraints. After the optimization model is solved, we can calculate the estimated maximum frequency deviation $\Delta f_{\text{EM}}^{i,t}$ of each grid

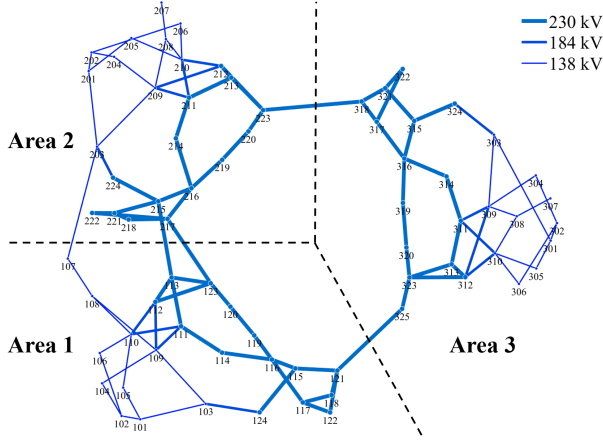


Fig. 7. The geographical layout of the RTS-GMLC test system.

and each time period by (68). The actual maximum frequency deviation $\Delta f_{AM}^{i,t}$ can be obtained using more detailed models or time domain simulations.

$$\Delta f_{EM}^{i,t} = \frac{\Delta P^i}{\min \left\{ \alpha_{(1)}^T X^{i,t} + \beta_{(1)}, \dots, \alpha_{(K_{pw1})}^T X^{i,t} + \beta_{(K_{pw1})} \right\}} \quad (68)$$

We denote $\Delta f_{AM}^i = \max_t \Delta f_{AM}^{i,t}$ and $\Delta f_{EM}^i = \max_t \Delta f_{EM}^{i,t}$. The difference between Δf_{AM}^i and Δf_{EM}^i reflects the error of the frequency nadir constraints. If Δf_{AM}^i is greater than the pre-defined frequency security limit Δf_{max}^i , we can modify the frequency security limit:

$$\Delta \hat{f}_{max}^i = \Delta f_{max}^i - (\Delta f_{AM}^i - \Delta f_{EM}^i) \quad (69)$$

After modification, the error of the frequency nadir constraints is offset. Solve the optimization model again, and the frequency security limit can be strictly satisfied.

V. CASE STUDY

A. Case Setup

A modified version of the RTS-GMLC test system is chosen as the testbed. The RTS-GMLC test system, proposed in [33], is an updated version of the RTS-96 test system. Time series data of wind, solar, and load are also included in the RTS-GMLC test system. The geographical layout of the RTS-GMLC is shown in Fig. 7, which includes 74 buses, three areas, and five inter-area tie lines. In this study, these five inter-area tie lines are changed into HVDC links, so that the original test system is modified into three asynchronous areas.

The generation mix includes thermal, nuclear, hydro, PV, and wind, among which thermal and hydro units are assumed to provide primary frequency regulation service. For each asynchronous area, a generation loss equal to 10% area load is considered as the incident of power disturbance. By default, we set $R_g = 0.08$, $K_g = 1$, $F_g = 0.3$, $T_g = 8$ for all generators, $D = 1$ for all areas, and $R_{DC} = 0.05$ for all HVDC lines. For the dataset generation in Appendix A, we set $K_{max} = 1.5$, $K_{min} = 0.5$, $N_{dim} = 5$. For the convex piecewise-linear regression, we set the number of segments $K_{pwl} = 4$.

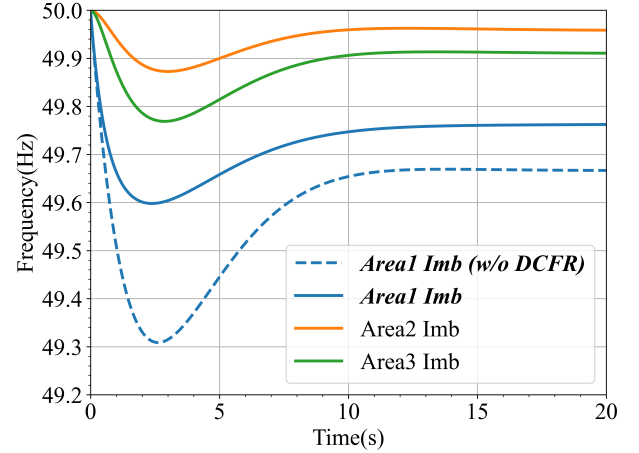


Fig. 8. The frequency response of Area 1 when the power disturbance occurs in different areas.

For the optimization models, we consider an hourly time resolution and a time horizon of 24 hours. All models are formulated in Python using Pyomo and then solved using Gurobi.

B. Numerical Analysis of Frequency Response Model

Using the SFR-MAG model described in Section II, the frequency response of each asynchronous area under the incident of power imbalance can be calculated. Fig. 8 shows the frequency response of Area 1 when the power imbalance occurs in different areas, where "Area 1 Imb" indicates that the power imbalance occurs in Area 1 and "w/o DCFR" indicates that HVDC frequency regulation is not enabled. When the power imbalance occurs in Area 1 itself and the HVDC frequency regulation is not enabled, the frequency drops to a nadir of 49.31Hz as the dashed curve. After enabling HVDC frequency regulation, the minimum frequency has an obvious improvement. When the power imbalance occurs in other areas, the frequency of Area 1 also drops because of the mutual influences caused by HVDC frequency regulation. The characteristics of the other areas are similar.

The comparison of maximum frequency deviations (corresponding to the frequency nadir) without and with DC frequency regulation is provided in Table I. It can be observed that the maximum frequency deviations significantly reduce after enabling the HVDC frequency regulation, especially for Area 3, where the frequency security problem is prominent. The comparison of maximum frequency deviations with power imbalance occurring in different areas is provided in Table II. When the power imbalance occurs in a certain area, the frequency deviation of itself is always the most severe, as marked in bold in the table. The inter-area influence is relatively small, and the frequency security risk of each area mainly results from the power imbalance of itself. Therefore, the power imbalances of other areas are not considered when adding frequency security constraints to the optimization model. Combining the above-mentioned results, it can be concluded that the frequency security problem of asynchronous grids can be greatly alleviated by introducing HVDC frequency regulation.

TABLE I
COMPARISON OF MAXIMUM FREQUENCY DEVIATIONS WITHOUT AND WITH DCFR

Area No.	Max frequency deviation (Hz)		Percentage reduction (%)
	w/o DCFR	w/ DCFR	
1	0.6912	0.4020	41.85
2	0.4877	0.3381	30.66
3	1.4756	0.7452	49.50

TABLE II
MAXIMUM FREQUENCY DEVIATIONS WITH POWER DISTURBANCE OCCURRING IN DIFFERENT AREAS

Area No.	Area 1 Imb	Area 2 Imb	Area 3 Imb
1	0.4020	0.1225	0.1525
2	0.1274	0.3381	0.1355
3	0.2311	0.1974	0.7452

The frequency nadir is largely affected by the HVDC frequency droop coefficients. Assuming the frequency droop coefficients of all the converters are the same, the relationship between the frequency droop coefficient and the frequency nadir of each asynchronous area is depicted in Fig. 9. When $\frac{1}{R_{DC}}$ is relatively small, the frequency nadir can be significantly improved by increasing $\frac{1}{R_{DC}}$. However, when $\frac{1}{R_{DC}}$ is relatively large, the marginal effect decreases. Here the physical meaning of $\frac{1}{R_{DC}}$ is the change in HVDC power under unit frequency deviation.

C. Precision of the Frequency Nadir Constraint Extraction

The performance of the approximate solutions obtained by the methods in Section III-A, i.e. the simplified method and the parameter aggregation method, is tested. To compare with related works, we consider the estimation method proposed in [26]. Assume a power imbalance event occurs in a certain area, and the maximum frequency deviations of the imbalanced area calculated by these three methods and the corresponding errors are provided in Table III. Scenarios with different HVDC

frequency droop coefficients are compared. The estimation method proposed in [26] need a preset maximum frequency deviation to approximate the frequency response of the grids neighboring to the imbalanced grid. When the actual maximum frequency deviation is less than the preset threshold (0.6Hz), the error is positive, and vice versa. The simplified method only considers the neighboring grids providing frequency support to the imbalanced grid and ignores the bilateral influence, thus has the largest error. The parameter aggregation method uses a set of aggregated parameters to approximate the bilateral influence between the imbalanced grid and the neighboring grids. The parameter aggregation is strictly equivalent only if the ratios among the parameters are exactly the same for all the neighboring grids. This condition is not satisfied for the test system, which causes the error of the parameter aggregation method. The percentage errors of all the three methods increase as $\frac{1}{R_{DC}}$ increases. All the three methods make some simplifications in the coupling between the imbalanced grid and the neighboring grids. Therefore, when the coupling becomes stronger ($\frac{1}{R_{DC}}$ increases), the errors of all the three methods increase. Among them, the parameter aggregation method has the highest accuracy. It is also worth mentioning that the errors of both the parameter aggregation method and the simplified method are always negative, which means that the maximum frequency deviations of these two methods are smaller than that of the precise method.

Next, the precision of the convex piecewise-linear regression described in Section III-B is validated as follows. Because the parameter ranges of the asynchronous areas are different, the process of generating dataset and convex piecewise-linear regression is implemented for each area respectively. The dataset for each area is generated using the parameter aggregation method and is divided into the training dataset and the testing dataset. Mean absolute percentage error (MAPE) is applied to measure the error. The MAPEs in the testing dataset for the three areas are 0.8078%, 0.5043%, and 0.7681% respectively. The testing MAPE is less than 1% for any area, supporting the precision of the convex piecewise-linear regression method.

D. Optimization Results

Five different optimization models are compared to demonstrate the benefits of the proposed model:

- 1) *Base*: the frequency constraints are ignored and the HVDC frequency regulation is not enabled.
- 2) *FC*: the frequency constraints are considered for each asynchronous area and the HVDC frequency regulation is not enabled.
- 3) *FC-cDC*: the HVDC frequency regulation is enabled and the frequency constraints are generated from the conservative frequency constraint generation method proposed in [26].
- 4) *FC-sDC*: the HVDC frequency regulation is enabled and the frequency constraints are generated from the simplified method.
- 5) *FC-DC*: the HVDC frequency regulation is enabled and the frequency constraints are generated from the parameter aggregation method.

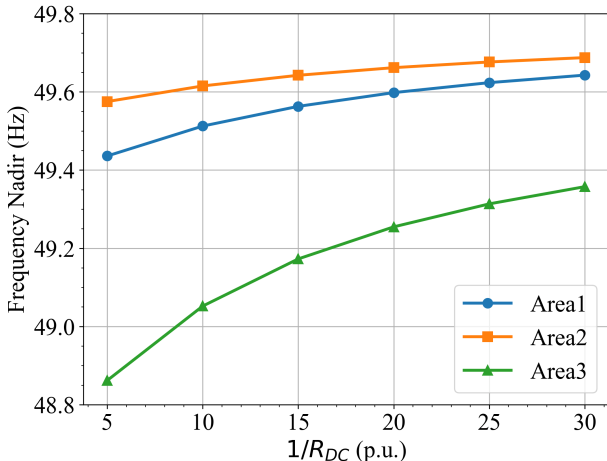


Fig. 9. The relationship between the frequency droop coefficient and the maximum frequency deviations.

TABLE III
THE COMPARISON OF THE MAXIMUM FREQUENCY DEVIATIONS CALCULATED BY THE DIFFERENT METHODS.

Area No.	$\frac{1}{R_{DC}}$	Max frequency deviation (Hz)			Percentage error		
		Estimation method [26]	Simplified	Parameter aggregation	Estimation method [26]	Simplified	Parameter aggregation
1	10	0.4996	0.4614	0.4857	2.48%	-5.35%	-0.37%
	20	0.4298	0.3442	0.3921	6.91%	-14.37%	-2.46%
2	10	0.3988	0.3655	0.3823	3.60%	-5.04%	-0.66%
	20	0.3685	0.2911	0.3268	8.96%	-13.92%	-3.36%
3	10	0.9329	0.8967	0.9420	-1.57%	-5.39%	-0.62%
	20	0.7190	0.6396	0.7212	-3.52%	-14.16%	-3.23%

These models have the same objective function and steady-state operational constraints, except that the frequency constraints and the HVDC frequency regulation reserve constraints are different. Note that the frequency constraint modification approach has not been applied yet. Based on the unit commitment solutions, the maximum frequency deviation of each area and each time period is calculated by the SFR-MAG model, as shown in Fig. 10. The operational cost and the maximum frequency deviation in the worst situation of each model are also annotated in Fig. 10. The Base model has the lowest cost. However, it does not enforce any frequency constraints and the maximum frequency deviation significantly exceeds the security limit. Area 3, whose local frequency regulation resources are the least, has the most prominent frequency security risk. The FC and FC-sDC models can maintain frequency security, although there are still errors in the frequency constraints. The operational cost of the FC model is the highest because the frequency security constraints are satisfied using only local frequency regulation resources. Taking advantage of HVDC frequency regulation, the FC-cDC, the FC-sDC, and the FC-DC can constrain the maximum frequency deviation with relatively low costs. Among these three models considering HVDC frequency regulation, the cost of the FC-cDC model is the highest due to the conservative frequency constraints. The frequency constraints of the FC-sDC model and the FC-cDC model have much larger errors compared with the FC-DC model. Therefore, the FC-DC model can better balance the operational cost and the frequency security.

E. Results of the Full-Order SFR Model and Frequency Constraint Modification

For the frequency constraint modification, a full-order SFR model is constructed with the IEEE G1 governor-turbine model based on Simulink. The detail description of the IEEE G1 governor-turbine model can be found in [31]. We set $T_1 = 0.2, T_2 = 0, T_3 = 0.1, T_4 = 0.2, T_5 = 8, T_6 = 0, T_7 = 0, K_1 = 0.3, K_3 = 0.7, K_5 = 0, K_7 = 0$ for the IEEE G1 governor-turbine model.

Based on the unit commitment solution, the full-order SFR model is implemented. The comparison of the SFR-MAG and the full-order SFR simulation results is provided in Fig. 11, which is corresponding to the unit commitment solution of the FC-DC model at hour 14. Though the frequency nadir calculated by the SFR-MAG model is constrained within the

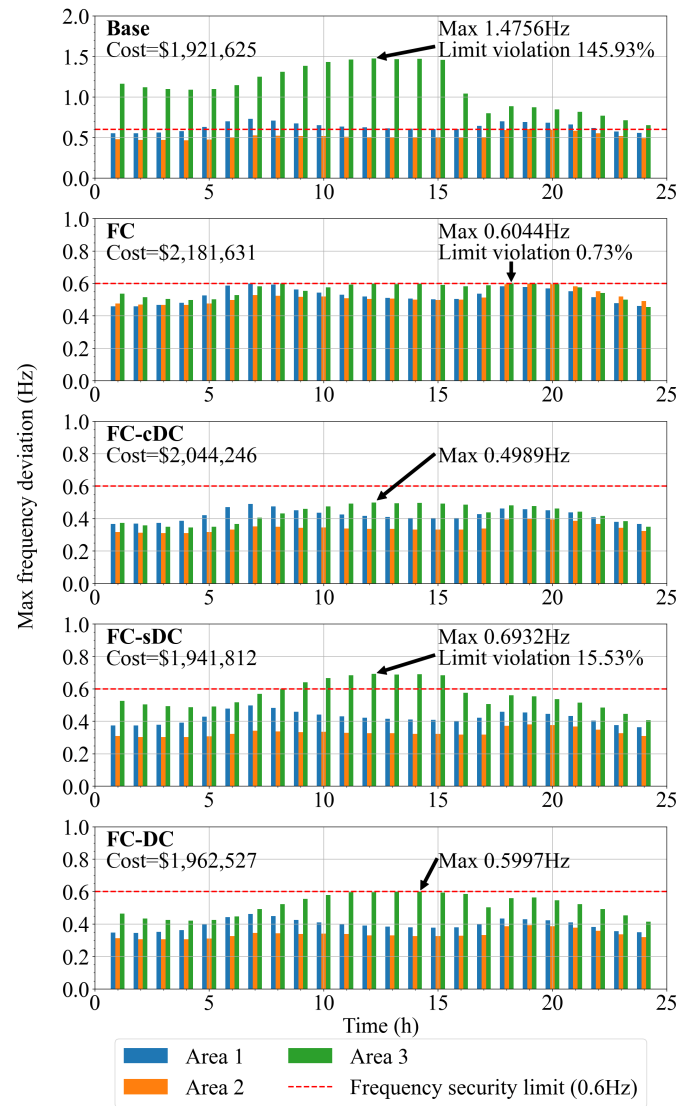


Fig. 10. The optimization results of different models.

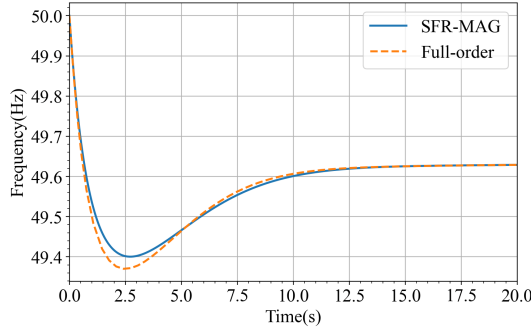


Fig. 11. Comparison of the SFR-MAG and the full-order SFR simulation results.

TABLE IV
THE PERFORMANCE COMPARISON OF FC-sDC AND FC-DC WITH THE FREQUENCY CONSTRAINT MODIFICATION APPROACH

	$\frac{1}{R_{DC}}$	Operational cost (\$)		Max frequency deviation of simulation (Hz)	
		FC-sDC	FC-DC	FC-sDC	FC-DC
Before modification	10	1,983,092	1,995,469	0.6695	0.6372
	20	1,941,812	1,962,527	0.7271	0.6298
After modification	10	2,053,354	2,039,248	0.5913	0.5968
	20	1,979,812	1,974,589	0.5888	0.5959

security limit, the result of the full-order SFR simulation exceeds the limit. The error of the frequency nadir is 4.8%.

Next, the frequency constraint modification approach is applied using the full-order SFR simulation results, and the performance of FC-sDC and FC-DC is further examined, which is shown in Table IV. The maximum frequency deviations of the simulation results of both FC-sDC and FC-DC can be constrained within the security limit after modification. The operational cost of FC-DC is consistently lower than that of FC-sDC, which can be explained that the frequency security boundary of FC-DC is more accurate.

VI. CONCLUSION

Enabled by HVDC frequency regulation, the primary frequency reserve resources can be shared among multi-area asynchronous grids, which brings significant operational benefits. This work studies how to model the system frequency response and extract the frequency constraints for the operation of MAG. A system frequency response model for MAG is developed to calculate the frequency response of each asynchronous area considering the mutual frequency support. An approximation algorithm to extract frequency constraints is proposed, which contains two steps: obtaining an approximate solution and convex piecewise-linear regression. Each step is validated to be highly accurate. Embedded with the frequency constraints, a frequency-constrained unit commitment model for MAG is proposed. A frequency constraint modification approach is introduced in the FCUC model to guarantee that the frequency security limit can be strictly satisfied even if there are errors in the fitted expression of the frequency constraint. The case study illustrates that the optimization model can

maintain the frequency security of multi-area asynchronous grids as well as exploit the operational benefits of HVDC frequency regulation.

APPENDIX

LINEARIZATION APPROACH FOR FRACTIONAL TERMS

Assume a fractional term of binary variables, where a_i is non-negative coefficients and u_i is binary variables:

$$y = \frac{1}{\sum a_i u_i} \quad (70)$$

It can be regarded as a non-linear constraint:

$$\sum a_i u_i y = 1 \quad (71)$$

By introducing non-negative auxiliary variables $z_i = u_i y$, we can transform (71) into the following linear constraints, where K is a constant greater than y :

$$\sum h_i z_i = 1 \quad (72)$$

$$y - z_i \leq K - K u_i \quad (73)$$

$$z_i \leq y \quad (74)$$

$$z_i \leq K u_i \quad (75)$$

DATASET GENERATION FOR CONVEX PIECEWISE-LINEAR REGRESSION

A small dataset of the unit status is generated by solving the unit commitment model without frequency nadir constraints. In practice, this small dataset can be drawn from the system historical data. This step aims to estimate a rough range of the parameters. For AC Grid i , its parameter vector $\mathbf{X}^{i,t}$ at each time period t can be calculated by (60)-(67). Moreover, the average relative error ϵ of the parameter aggregation method can be calculated on this small dataset.

For the k^{th} parameter x_k^i , we choose the maximum value $x_{k,\max}^i$ and the minimum values $x_{k,\min}^i$ in the small dataset. We set $\bar{x}_k^i = K_{\max} x_{k,\max}^i$ and $\underline{x}_k^i = K_{\min} x_{k,\min}^i$ as the upper and the lower bounds of this parameter, where $K_{\max} \geq 1$ and $K_{\min} \leq 1$ are constants that determine the parameter range of the regression dataset. By evenly dividing the range of the each parameter into N_{dim} and listing all the possible combinations of these parameters, we can generate totally $N_{\text{data}} = (N_{\text{dim}})^8$ points of input data. Note that the function $\frac{1}{h(\mathbf{X}_n^i)}$ is the calculation result of the parameter aggregation method, which still has error. Therefore, we calculate $y_n^i = \frac{1}{(1+\epsilon)h(\mathbf{X}_n^i)}$ as the output data to compensate the error of the parameter method and further improve the accuracy of the frequency constraint extraction. By this means, the dataset $(\mathbf{X}_n^i, y_n^i)$, $n = 1, \dots, N_{\text{data}}$ is generated.

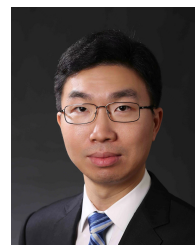
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